ACADEMIC CURRICULUM VITAE

Pedro Porras Flores

Ph.D. in Mathematics

UNAM / Universitat de Barcelona

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Academic profiles

ORCID: 0009-0009-4624-0863 | Google Scholar | ResearchGate | GitHub

Education

Ph.D. in Mathematics Feb. 2020 – Nov. 2025

UNAM / Universitat de Barcelona (cotutelle).

Thesis: *Constructive methods in KAM theory for quasi-periodic time-dependent Hamiltonian systems and applications*. *Cum laude* (2025) and International Doctorate Mention (2026), Universitat de Barcelona.

M.Sc. in Mathematics Jan. 2017 – Dec. 2018

UNAM. Graduated with honors.

B.Sc. in Physics Aug. 2008 – Dec. 2013

UNAM.

Professional experience

Academic Technician (Técnico Académico Asociado "C"), full time Jun. 2025 – present

DGTIC (Directorate of Computing and Information Technologies), UNAM.

Software and database test automation; performance analysis and validation of computing systems according to quality standards; technical documentation and configuration control.

Adjunct Lecturer (Profesor de asignatura A) 2015 – present (non-continuous periods)

UNAM.

Faculty of Sciences: Aug. 2015 – Aug. 2021; Aug. 2022 – Aug. 2023; semesters 2026-1 and 2027-1 (ongoing).
National School of Earth Sciences: Aug. 2020 – Feb. 2022.

Teaching Assistant (Ayudante de asignatura B) 2013 – 2021 (non-continuous periods)

UNAM.

Faculty of Sciences: Jan. 2013 – Aug. 2015; Feb.–Aug. 2016; Jan.–Aug. 2017; Sep. 2020 – Feb. 2021.

Research projects

PAPIIT–DGAPA, UNAM.

Projects led by Dr. Renato Carlos Calleja Castillo, Department of Mathematics and Mechanics, IIMAS-UNAM. The doctoral research was carried out within these research lines and with their academic support. Participation as project member:

- **IN101020.** Project activities, including research stays at the University of Barcelona and Uppsala University.
- **IN103423.** Project activities, including research stays at the University of Barcelona and Uppsala University.

- **IN104725.** Project activities, with support for completing the doctoral thesis, research stays at the University of Barcelona and conference attendance.

AEI PID2021-125535NB-I00 Feb. 2021 – Dec. 2024

Research project member, Universitat de Barcelona.

Analytic and Computational Methods in Dynamical Systems. Principal investigator: Dr. Alejandro Haro Provinciale. Study of non-autonomous Hamiltonian systems; formulation and proof of a posteriori KAM results; development and implementation of quasi-Newton algorithms for computing invariant tori. This work led to scientific publications with project members.

Research stays and international collaboration

- University of Barcelona, with Dr. Alex Haro: Jun.–Dec. 2024 and Nov. 2021 – Jul. 2022.
- Uppsala University, with Dr. Jordi-Lluís Figueras: Aug.–Nov. 2023.

Publications and work in preparation

Note on authorship. In mathematics, author lists are customarily given in alphabetical order. In both publications below, **Pedro Porras Flores is the corresponding author**, marked with †.

Articles in peer-reviewed journals

1. Renato C. Calleja, Àlex Haro and **Pedro Porras Flores**†. *Constructive approaches to QP-time-dependent KAM theory for Lagrangian tori in Hamiltonian systems*. *Journal of Differential Equations*, **449** (2025), 113681. doi:10.1016/j.jde.2025.113681.
2. Renato C. Calleja, Àlex Haro and **Pedro Porras Flores**†. *KAM algorithms for Lagrangian tori in time-quasi-periodic Hamiltonian systems*. *Communications in Nonlinear Science and Numerical Simulation*, **153** (2026), 109445. doi:10.1016/j.cnsns.2025.109445.

† Corresponding author.

Manuscripts in preparation

- Renato C. Calleja, Carlos García-Azpeitia and Pedro Porras Flores. *Numerical computation and continuation of Lagrangian invariant tori in the Sitnikov problem: an a posteriori KAM approach*. In preparation.
- Renato C. Calleja, Àlex Haro and Pedro Porras Flores. *Constructive KAM methods for partially hyperbolic tori in quasi-periodically time-dependent Hamiltonian systems*. In preparation.

Teaching materials and software. *Computational Physics 2027-1*: course materials, exercises and numerical tools in Python, public repository https://github.com/PPorras/Fisica_Computacional_2027-1.

Refereeing

Referee for *SIAM Journal on Applied Dynamical Systems*.

Talks and scientific communication

Selected conference talks, seminars and outreach activities.

- *Toros invariantes en sistemas cuasiperiódicos en el tiempo: del teorema KAM al algoritmo y las aplicaciones* (in Spanish). EMALCA, Yachay Tech University, Ecuador (online). International, invited. Jun. 20–31, 2026.
- *Constructive KAM Theory for Quasiperiodic Lagrangian Tori: Algorithms and Applications to the Sitnikov Problem*. Workshop “Hamiltonian Systems and Celestial Mechanics”, UNAM-Oaxaca, Mexico. International, invited. Jun. 7–12, 2026.

- *El problema de Kepler asimétrico* (in Spanish). Thesis Outreach Seminar, Faculty of Sciences, UNAM. May 28, 2026.
- *Methods in KAM Theory for Lagrangian Tori in Quasi-Periodically Time-Dependent Hamiltonian Systems*. 2025 SIAM Conference on Applications of Dynamical Systems (DS25), Colorado, USA. May 11–15, 2025.
- *Métodos Constructivos en Teoría KAM* (in Spanish). VI Joint Meeting RSME–SMM, Valencia, Spain. International, invited. Jul. 1–5, 2024.
- *Constructive KAM Methods for Lagrangian Tori*. Workshop on Computer-Assisted Proofs in Nonlinear Analysis, CRM Montreal, Canada. International, invited. Sep. 9–13, 2024.
- Poster: *Constructive Methods in KAM Theory for Quasi-Periodic Hamiltonian Systems with Application to Tokamaks* (with R. Calleja and A. Haro). DS23 – SIAM Conference on Applications of Dynamical Systems, Portland, USA. May 14–18, 2023.
- International Supercomputing Conference, UNAM, Mexico City. Sep. 1–5, 2025 (attendee).

Teaching and student training

Undergraduate courses — Faculty of Sciences, UNAM. Courses and semesters as recorded in the official teaching certificate (UNAM semester codes: *YYYY-1* = August–December of the previous year, *YYYY-2* = January–June).

As lecturer:

- *Computational Physics* (Physics): 2026-1.
- *Mathematics for Applied Sciences II* (Computer Science): 2023-2.
- *Computational Algorithms* (Biomedical Physics): 2018-2, 2019-2, 2020-2, 2020-4 (intersession) and 2023-2.
- *Mathematics for Earth Sciences I* (Earth Sciences): 2018-1, 2020-1 and 2023-1.
- *Mathematics for Earth Sciences II* (Earth Sciences): 2016-2, 2020-2 and 2021-2.
- *Mathematics for Earth Sciences III* (Earth Sciences): 2017-1, 2019-1 and 2021-1.
- *Computing* (Physics): 2016-1 and 2017-1.

As teaching assistant:

- *Mathematical Introduction to Celestial Mechanics* (Mathematics): 2026-2.
- *Computational Physics* (Physics): 2020-1 and 2021-1.
- *Collective Phenomena* (Physics): 2013-2 and 2017-2.
- *Advanced Mathematics for Physics* (Physics): 2016-2.
- *Electromagnetism I* (Physics): 2015-2.
- *Electromagnetic Phenomena* (Earth Sciences): 2014-1, 2014-2 and 2015-1.

Invited courses:

- *Numerical methods in KAM theory via the parameterization method*, Mexican HAT, UNAM (Dec. 2023).
- *The Taylor Integration Method in Dynamical Systems*, Summer School on Computational Simulation, UASLP (2023).

Supervision: three social-service students supporting teaching (Applied Mathematics/Earth Sciences, Faculty of Sciences, UNAM), Sep. 2021 – Apr. 2022.

Event organization:

- Co-organizer of the *Chaotic Seminar Dynamical Systems* (CRM Barcelona, ongoing since Oct. 2024).
- Co-organizer of the *TraX International Workshop* (CRM Barcelona, May 2022).

Juries:

- XXXI Science Outreach and Technological Innovation Project, Colegio de Educación Integral (Mar. 2026).

- 31st National Contest of Physics Devices and Experiments, Mexican Physical Society (2022).

Awards, honors and fellowships

- Recognition for 10 years of academic service at UNAM (Jun. 2025).
- *Cum laude* (2025) and International Doctorate Mention (2026), Universitat de Barcelona; master's degree with honors, UNAM (2019).
- SIAM travel award to attend DS23, Portland, USA (May 2023).
- MSRI travel award to attend the Hamiltonian Systems workshop, UC Berkeley (2018).
- CONACyT doctoral (Feb. 2021 – Jan. 2025) and master's (2018) fellowships.
- Member of the Society for Industrial and Applied Mathematics (SIAM), 2023–2025.

Additional training (selection)

- Computer-Assisted Proofs in Nonlinear Analysis, CRM Montreal (2024).
- Introduction to Data Science with Python and preparatory data science course, DGTIC-UNAM (2025).
- Specialized schools in dynamical systems at CRM Barcelona and BGSMath: GLADS 2022, JISD 2022, Hypatia Graduate Summer School 2022, BGSMath course on chaotic phenomena and invariant manifolds 2022, and Winter School DANCE 2022.

Technical skills and languages

Programming: Python, C++, Julia, SQL, PostgreSQL. **Tools:** Git/GitHub, Jupyter, Linux server administration.
Languages: Spanish (native); English (intermediate); Italian (working knowledge, spoken and written).